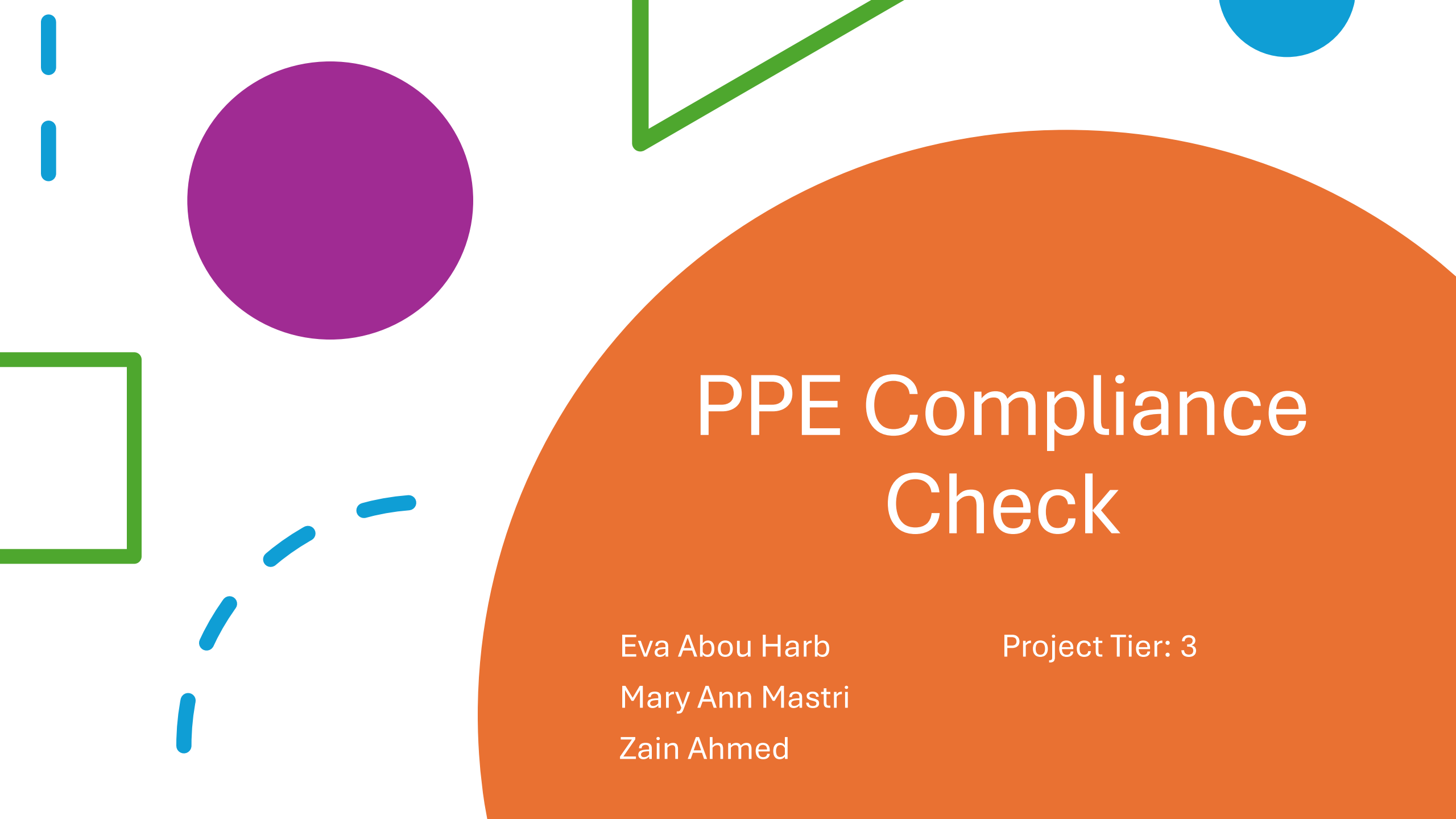




PPE Compliance Check

Eva Abou Harb
Mary Ann Mastri
Zain Ahmed

Project Tier: 3

The Problem

What real-world problem are we solving?

- Workers in industrial settings do not always wear the required personal protective equipment (PPE), increasing the risk of workplace injuries.

Who cares?

- Construction companies
- Safety managers
- Manufacturing facilities
- OSHA inspectors

Why is it important?

- **Reduce** workplace accidents
- **Improve** safety compliance
- **Lower** financial and legal risks



Our Solution

What will our system do?

- Detect workers and PPE in images
- Identify helmets, safety vests, and goggles
- Determine whether each worker is compliant

How will it solve the problem?

Image → YOLO11 → Detect PPE → Compliance Check



One-sentence description

An AI-powered system that automatically detects whether construction workers are wearing the required PPE using YOLO11 object detection.



Technical Approach

Computer Vision Technique: Object Detection

Model: YOLO11

Framework: PyTorch/Ultralytics

Dataset: Roboflow + Kaggle

- YOLO11 was selected because it provides fast and accurate real-time object detection, making it ideal for PPE detection in construction environments. PyTorch and Ultralytics provide an efficient framework for training, evaluating, and deploying the model.



Data Plan

Source:

- Roboflow Universe
- Kaggle PPE Datasets

Dataset Size:

- Roboflow dataset: 136 labeled images
- Kaggle dataset: 7,035 images

Labels:

- Person
- Hard Hat
- Safety Vest
- Safety Goggles

Data Preparation:

- Verify annotations
- Resize images
- Data augmentation
- Train / Validation / Test split



kaggle

System Diagram



Step 1 – Image Acquisition

Capture or upload an image of workers at a construction site.



Step 2 – Model Inference

YOLO11 processes the image and detects workers and PPE.



Step 3 – PPE Detection

Identify helmets, safety vests, and safety goggles.



Step 4 – Compliance Decision

Determine whether each worker is compliant or non-compliant.



Step 5 – Output & Visualization

Display detection results with bounding boxes, labels, and compliance status.



Success Metrics

Metric Type	Example	Target
Primary Metric	Accuracy, Precision, Recall	$\geq 90\%$ overall accuracy
Secondary Metric	Inference Speed (Latency)	$< 1\text{s}$ per image

Week-by-Week Plan



Task	Milestone
Get dataset & set up environment	✅ Dataset Ready
Train or fine-tune model	🤖 Model Working
Test and improve	📊 Good Accuracy
Create demo or video	🎥 Demo Ready
Final testing & documentation	📄 Everything Done
Present project	🎉 Presentation Day

Challenges & Backup Plans

 Challenge	 Plan B
Limited training data	Use additional Roboflow and Kaggle datasets.
Model accuracy below 90%	Apply data augmentation or use a larger YOLO11 model.
Poor lighting or image quality	Collect more diverse images and improve preprocessing.
Small or partially hidden PPE	Fine-tune the model with additional labeled images.

Resources Needed



Resource	Options / Notes
Compute	Google Colab, Kaggle
Frameworks	PyTorch, Ultralytics
Dataset Sources	Roboflow, Kaggle
Development Tools	Python, Jupyter Notebook
Estimated Cost	\$0 (using free educational resources)